Samarth Kumar

Final Year Undergraduate pursuing a double major in **Physics** and **Statistics & Data Science** with a minor in **Machine Learning**

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.S. (Double Major in SDS and PHY)	Indian Institute of Technology Kanpur	9.3/10
2021	Class XII (CBSE)	Gyanshree School, Noida	97.4%
2019	Class X (CBSE)	Gyanshree School, Noida	96.2%

Awards & Achievements

- Received the **Academic Excellence Award (2022-2023, 2023-2024)** for being amongst the **top 10**% of the batch. (*2024*)
- Received the **Haldiram Merit Scholarship** four years in a row for continued academic excellence. (*2021-2025*)

Work Experience

**: Incoming Quantitative Researcher at OpenFutures*

Quantitative Researcher - Intern | Edelweiss Global Markets (*May '25 – July '25*)

Objectives	• Refine mid-frequency strategies by learning latent market features for optimal entry and exit positions. • Create a smart strategy selector by expanding on the above model and incorporating multiple strategies.
Approach	• Analysing and creating market features and indicators that are able to capture various trends effectively. • Training **semi-supervised models** on labels created using look-forward approach on the training data. • Creating a rolling backtesting pipeline to locally compare model **returns, Sharpe and Calmar ratios**.
Results	• Created a model capable of optimising entry and exit timings to improve Sharpe and Calmar ratios. • Integrated and deployed model into existing frameworks for directional and non-directional strategies.

Quantitative Machine Learning Engineer - Intern | Blockhouse Labs (*December '24 – March '25*)

Objectives	• Develop a complete **back-testing simulator** that includes **market impact** effects for effective testing. • Enhance simulator to be able to cater for **multi-asset, multi-exchange order routing strategies**.
Approach	• Creating a **dynamic order book simulator** using a stochastic state-based model for live simulations. • Using historical market data as initial conditions for the simulator and the training of model parameters. • Implementing various **strategic indicators** to quantify strategies, and setting benchmark thresholds.
Results	• Created a dynamic **back-testing simulator**, capable of multi-order, multi-exchange order simulations. • Started process for setting benchmarks for **Smart Order Routing Strategies** and optimising them.

Quantitative Researcher - Intern | OpenFutures * (*May '24 – July ' 24*)

Objectives	• Analyse financial market data to understand market microstructure for mid-frequency trading strategies • Create data processing libraries and computational modules to be used for specialised data analysis.
Approach	• Developed an **Option Chain**, including calculations for **greeks and volatilities**, for NSE listings. • Formulated a real time multidimensional **Kalman Filter** in C++ using the **Eigen template library**. • Wrote an **analysis pipeline** in python and used it to study the **predicting power** of various indicators.
Results	• Automated the option map generation procedure, facilitating a streamlined process for further analysis. • Created modules that have been integrated into real-time mid and high-frequency trading strategies. • Studied the mean-reverting nature of volatility of returns in various financial assets and derivatives.

Research Experience

Summer Research Fellowship | RDS Labs IIT Roorkee (*May '23 – Present*)

Objectives	• To simulate a gamma radiation detector array for the purpose of researching efficiency corrections.
Approach	• Closely modelled the **4π icosahedron geometry** of the gamma ray detector array using Solidworks. • Employed the C++ based OOP toolkit **GEANT4** for particle simulations and efficiency calculations. • Utilised **CERN-ROOT** for data visualisation and created a data analysis module for corroboration.
Results	• Successfully attained the estimated efficiencies for 511keV and 1275keV emission of ^{22}Na radio-isotope.

Publications

- **Kumar, S.**, Gupt, N., Ghosh, S., Ghosh, A. Introduction to quantum thermodynamic cycles. J Chem Sci 135, 36.
  - The paper is an introduction to **Quantum Thermodynamics**, an important aspect in designing quantum technologies.
  - It reiterates the key aspects of classical thermodynamics and explains their **quantum analogues** in an intuitive manner.
  - Emphasis is placed on quantum cycles and their efficiency calculations, along with possible areas of **quantum supremacy**.

Competitions

- **Quantum Computing Challenge** | Tata Consultancy Services (*Dec'23 - March'24*)
  - Led a team in creating a novel algorithm, leveraging **qGANs** in an innovative approach for stocks price prediction.
  - Aided in the creation of a **modified QUBO** method to address the tail assignment problem of the airline industry.

Key Projects

- **Curve Point Identification** - Course Project | Dr. Dootika Vats, IITK (*Feb '25*)
  - Extracted **geometric features** such as turning angles, curvature, slope, and distance between points from the coordinate data.
  - Trained a **Random Forest classifier** to distinguish curved road segments, achieving reproducible results over various precisions.
- **Era-Adjusted Cricket Analytics** - Course Project | Dr. Dootika Vats, IITK (*Jan '25*)
  - Built a pipeline to classify players into cricketing eras and normalize statistics using **mean-scaling** and **Z-score standardization**.
  - Designed a linear **rank statistic** combining runs, average, and strike rate, producing reproducible, era-adjusted player rankings.
- **Low Cost Gamma Ray Detector** - Course Project | Dr. Aditya H. Kelkar, Dr. Krishnacharya, IITK (*Sep'23 - Nov'23*)
  - Led a team in developing a cost-effective **gamma-ray spectrometer** using a PMT, basic circuit elements and an Arduino UNO.
  - Engineered a data acquisition system by creating a **peak detector circuit**, and then implemented real-time spectrum analysis.
- **Computational Genomics** - Course Project | Dr. Hamim Zafar,IITK (*July'23 - Nov'23*)
  - Contributed to the development of a **multi-modal variational auto-encoder**, using generative models to learn cell embedding.
  - Performed specialised analysis including clustering, dimensionality reduction, imputation, sequencing, and multi-omics integration.
- **Rainfall Prediction** | Dr. M. K. Verma,IITK (*July'23 - Nov'23*)
  - Implemented a **supervised deep learning model** for rainfall forecasting in Indian cities using ten-year airport weather data.
  - Created a **grid search algorithm** to optimize the model hyper parameters in an effort to increase accuracy of neural network.
- **Translational Neuroscience** | Dr. Pragrathi Balasubramani, IITK (*September '22 - July'23*)
  - Executed a statistical analysis of the effects of psychiatric medication's cognitive and behavioural effects on mental health patients.
  - Designed an analysis pipeline on a drug study, employing **statistical tests** to understand its effects on interventional outcomes.
- **Simulated Day Trading** | Finlatics Market Experience Program (*Jan'22 - April'22*)
  - Achieved a **100% score** in the equity trading simulator by efficiently managing the trading portfolio for the complete duration.
  - Conducted small scale market research on various companies in the pharmaceutical and technological sectors to optimise trades.

Technical Skill

- **Python Libraries and Toolkits**: PyTorch, TensorFlow, Numpy, Pandas, Scikit-learn, statsmodels, Qiskit
- **C/C++ Toolkits and Frameworks**: FLTK, Cern-GEANT4, Cern-ROOT, BOOST
- **Other Tools**: R, Git

Extra-Circulars

- **Quantum Information Reading Group** | Science Coffee House,IITK (*April'23 - May'24*)
  - Contributed to a student-led reading group dedicated to exploring theoretical quantum computing and information.
  - Engaged in discussions encompassing diverse topics such as entanglement, the CHSH game, and Bell states.
- **Stamatics** | Mathematics and Statistics Society,IITK (*June'22 - July'22*)
  - Contributed to a student-led project aimed to provide an introduction to higher mathematics, emphasising Geometrical Mechanics.
  - Studied and formulated an introductory article on the topic of 'Non-Orient-able Surfaces' emphasising the Mobius strip.
- **Working Model Droid** | Robotics Club,IITK (*May'22 - August'22*)
  - Supported the development of a working replica of the 'BB8' droid from Star Wars, utilising an Arduino Mega board.
  - Added controlled navigation and improved control capabilities by implementing Bluetooth connectivity to the droid.

Positions of Responsibility

- **Secretary** | Music Club,IITK (*May'22 - August'23*)
  - Assisted in the organisation of the bi-annual even **Musical Extravaganza 2023**, the flagship event of the music club.
  - Conducted a month long workshop, mentoring individuals from various domains in the field of music production.

Relevant Coursework

:Excellent Performance* *:Outstanding Performance*

Statistics, Data Science, and Machine Learning

Stochastic Processes*	Fundamentals of Computing	Linear Algebra
Multivariate Analysis**	High Performance Computing	Time Series Analysis**
Statistical Computing**	Data Structures & Algorithms	Probability & Statistics*
Techniques in Data Mining**	Introduction to Machine Learning**	Linear Regression and ANOVA**

Physics and Engineering

Optics*	Relativity*	Classical Mechanics**
Thermal Physics*	Statistical Mechanics*	Classical Electrodynamics*
Thermodynamics	Computational Physics**	Modern Physics Laboratory
Experimental Physics	Fluid Mechanics and Rate Processes*	Quantum Mechanics I, II, III *

Licenses and Certifications

Quantum Information | **Quantum Computing** | **Machine Learning**